Sprint 4 Team Project assignment

I am using multiclass logistic Regression; Softmax Cross entropy to approach this problem.

- Here multiple features are involved in decision

- The amount to call
  - the pot
  - player's remaining stack
  - pot odds
  - whether player faces a bet
  - raises so far
  - players still in
  - position
  - flop, turn

This model should predict fold, call/check or raise - Goal to model the probability of each class given the situation

Here are the notations I plan to use

N = number of decision in the full dataset

D = Number of features,
~~including~~ this will be D+1 including bias

K = 3 classes to consider

3 classes $\{C_0 = \text{fold}, C_1 = \text{call}, C_2 = \text{raise}\}$

The we will describe each decision with a feature vector.

after that we will assign linear scores one per class.

and softmax turns scores into probalities

~~and~~ then we get likelihood and have cross entopy of negtive likelihood

then we try to find gradient and then gradient descent to find global minimum.